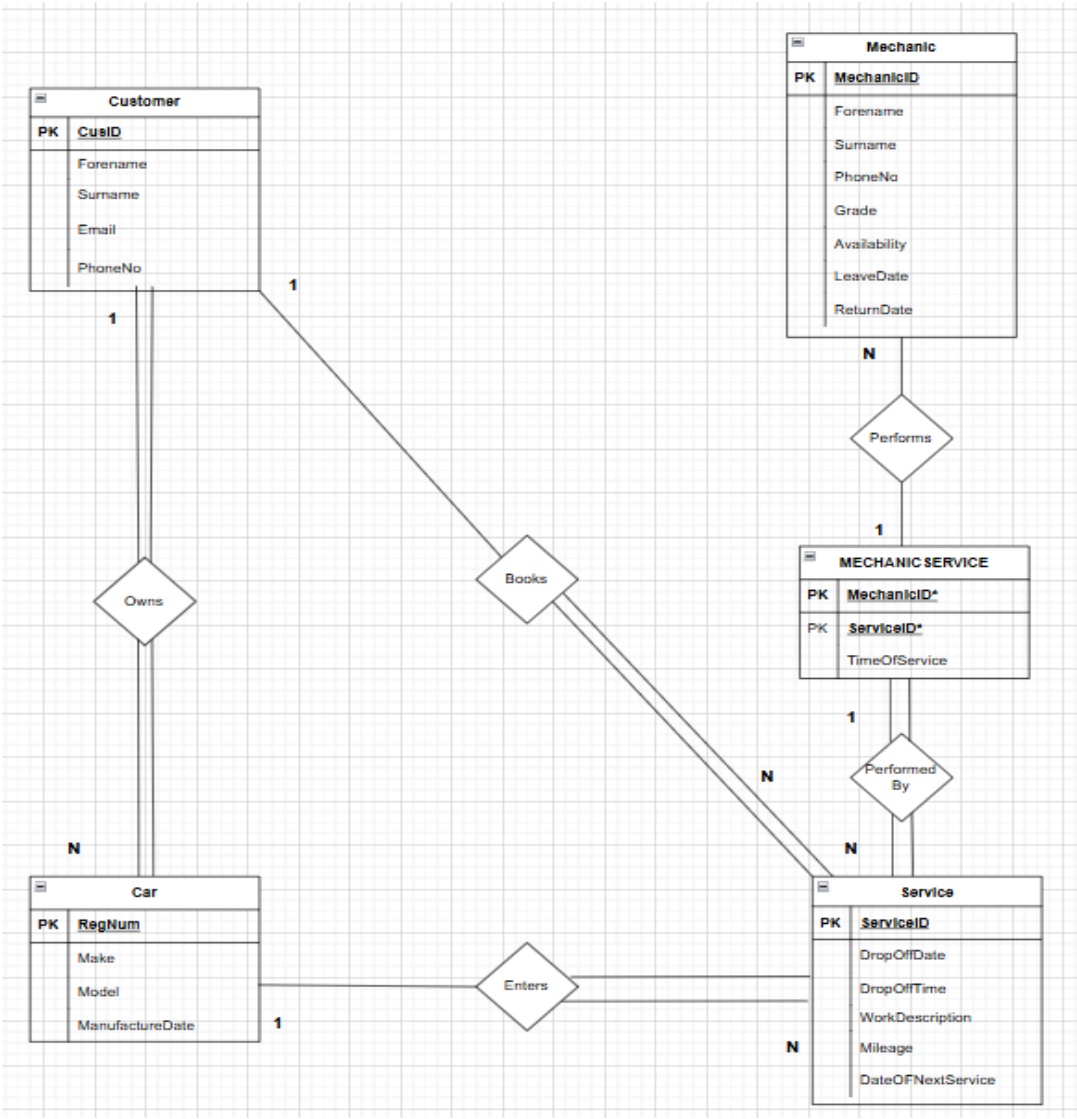



Customer-Car: Full participation from both sides. A car must be owned and booked in be a customer and a car can't have no owner and have a booking made. It is a 1 to many relationship. All customers can own and book in multiple cars but a car that's booked in will only be under the name of 1 owner

Car-Service: Full participation from service to car but partial from car to service. Each service must have a car but a car may be kept on the database and not have another service booked (similar scenario to the customer. It is 1 to many relationship from car to service. A car can be booked in for multiple services throughout a period but each service record will only have one car it's associated with.

Service-MechanicService: It is full particicipation. A service can't be completed without a mechanic/mechanics working on it so it's impossible for completion of a service if there's no mechanic. It is a 1 to many relationship as 1 service may be worked on multiple mechanics throughout its completion.

Mechanic-MechanicService: This is a partial participation. A mechanic may either not be available or may be in training so doesnt have a service booked at all. So not every

mechanic needs to be working on a service. It is a 1 to many as a mechanic can have many services booked over a period (e.g an MOT check up on monday and annual service on wednesday, both really should be in the database in preperation for the booking.)

MechanicService is a link table to bring together the many to many relationship of service and mechanic. A mechanic can work on multiple services and multiple services can be worked on by a mechanic

Customer-Service: Service to customer is full participation but customer to service is partial participation. Each service is booked by a customer or else it isn't created but a customer may be in the database and not have a service booked. While DateOfNextBooking does exist, there is no guarantee that is automatically added to the database straight away so a car can go without a service for a while. The relationship is 1 to many as a car can go into multiple different services but a service booking will only focus on 1 car

## Relational Schema

Basic Tables:

CUSTOMER(**CusID**, Forename, Surname, Email, PhoneNo)

CAR(**RegNum**, Make, Model, Manufacture)

SERVICE(**ServiceID**, DropOffDate, DropOffTime, WorkDescription, Mileage, DateOfNextService)

MECHANICSERVICE(**ServiceID, MechanicID,** TimeOfService)

MECHANIC(**MechanicID**, Forename, Surname, PhoneNo, Grade, Availability, LeaveDate, ReturnDate)

Final Relational Schema:

CUSTOMER(**CusID**, Forename, Surname, Email, PhoneNo)

CAR(**RegNum**, Make, Model, Manufacture, CusID*)

SERVICE(**ServiceID**, DropOffDate, DropOffTime, WorkDescription, Mileage, DateOfNextService, RegNum* CusID*)

PERFORMS(**ServiceID*, MechanicID***, TimeOfService)

MECHANIC(**MechanicID**, Forename, Surname, PhoneNo, Grade, Availability, LeaveDate, ReturnDate)

# Assumptions and Limitations

Assumptions:

- A customer can be registered on the database without being linked to a service (e.g if they have recently had a service completed but may use the company again).
    - They may have used the service but havent booked a follow up appointment or wish to do so at a later point

- Each service is assigned to a mechanic (as this would be needed to decide the time of the service)
    - This would be required to decide the time of the service as it would be risky to take the booking if you dont know for sure that there will be an available slot.

- Not every mechanic is assigned to a service
    - Could be:
    - New mechanic
    - In training (due to being new or learning a new service)
    - On a long period of leave and no services have been assigned

- All Cars must be associated with an owner

- A car is associated with only 1 owner at a time

- A mechanic only works on 1 service at a time

Limitations:

- The same car can have a different owner but will still be under the original owner's ID at least temporarily
- No way to list what has actually been done. The WorkDescription only lists work that WILL be done. Any extra changes or if anything changes, it's not noted.
- There is no tracking of any changes made to the car (e.g if the registration number changes)
- Unless customer brings it up, if there is any changes to the customer contact details then it may not be possible to contact the customer.
- No tracking of possible delays to jobs which could complicate assigning mechanics to their next jobs
- Could be confusing if a car is brought in by a different person that isn't the owner (e.g a spouse or friend).

- Mileage of the car can change significantly between the booking of a service and when it happens.
- No tracking of when mechanics takes breaks and how long they'd be for.